

Separable Differential Equations



Solving by separation of variables

- 1 Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
 - 2 Solve $\frac{dy}{dx} = 2xy$, given $y(0) = 5$.
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Growth, decay, and applications

- 3 A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.
 - a Solve for $P(t)$.
 - b Find $P(20)$, to the nearest whole number.
 - 4 Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a 20°C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.
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More separable equations

- 5 Solve $\frac{dy}{dx} = \frac{y}{x}$, given $y(1) = 4$.
 - 6 Solve $\frac{dy}{dx} = y(1 - y)$ (logistic-type equation), given $y(0) = \frac{1}{2}$, using partial fractions $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.
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Mixing and applied models

- 7 A tank has 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution drains at the same rate, so the volume stays constant. Let $y(t)$ be the amount of salt (kg) at time t . The differential equation is $\frac{dy}{dt} = 6 - \frac{3y}{100}$.
 - a Find the equilibrium value of y (where $\frac{dy}{dt} = 0$).
 - b Explain, in words, what this equilibrium represents physically.
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Slope fields and qualitative behavior

- 8 For the differential equation $\frac{dy}{dx} = x - y$, find the slope of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$.

- 9 Verify that $y = x - 1 + 2e^{-x}$ is a solution to $\frac{dy}{dx} = x - y$, by substituting into both sides.